

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 20%

QUESTIONS: 05 Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

<i>Criteria</i>	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	<i>Total(10)</i>
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<i>Weightage</i>	<i>20%</i>	<i>25%</i>	<i>20%</i>	<i>20%</i>	<i>10%</i>	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code

of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

*I understand that any violation of academic integrity rules will result in disciplinary action. **Signature***

of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant

Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes

Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

Customer Income (₹) Spending Score

C1 30000 60

C2 60000 40

C3 50000 70

C4 28000 65

C5 65000 35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

Q2. Dataset: Document Data

Document Word1 Word2 Word3

D1 0.2 0.5 0.3

D2 0.1 0.6 0.3

D3 0.8 0.1 0.1

D4 0.75 0.15 0.1

D5 0.2 0.4 0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. (BL4 – 1 Mark)

Q3. Dataset: Geographic Data

Point Latitude Longitude

P1 13.08 80.27

P2 13.10 80.25

P3 12.97 80.22

P4 13.05 80.30

Point Latitude Longitude

P5 13.00 80.20

Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)

Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1 50000 20

P2 30000 10

P3 52000 22

P4 25000 8

P5 32000 12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)

Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1 500 A 4.5

I2 300 B 3.8

I3 550 A 4.7

I4 250 B 3.5

I5 520 A 4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)

Q1 - Answer

Cluster mode

☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
(Num) SpendingScore
☒ More clusters for visualization

Ignore attributes

Start Stop

Result for (right-click for options)

06-44-01 - SimpleKMeans

06-40-07 - SimpleKMeans

Cluster output

*** Clustering model (full training set) ***

KMeans

Random of iterations: 2

Within cluster sum of squared errors: 0.27436197476167916

Initial starting points (random):

Cluster 0: 0,45

Cluster 1: 0.949195,60

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster0	Cluster1
	(5,0)	13,01	02,00
Income	0.3007	0.2282	0.3324
SpendingScore	04	45	27,5

Time taken to build model (full training data) : 0 seconds

*** Model and evaluation on training set ***

Clustered Instances

0 3 (60%)

1 2 (40%)

Cluster mode

☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
(Num) SpendingScore
☒ More clusters for visualization

Ignore attributes

Start Stop

Result for (right-click for options)

06-44-01 - SimpleKMeans

06-46-00 - SimpleKMeans

06-47-02 - HierarchicalCluster

Cluster output

*** Run Information ***

Jobname: mka-clusters-mk-hierarchical-clusters -W 1 -L SINGLE -P -k "mka.csv.EuclideanDistance -B Hier-Inst"

Scriptfile: customers*mka-filters-unsupervised.attribute-normalize-01-0-00-0*mka-filters-unsupervised.attribute-remove-01

Instances: 5

Attributes: 3

Income

SpendingScore

Test mode: evaluate on training data

*** Clustering model (full training set) ***

Cluster 0

(00,0+0.35276,45,0+0.35276)+0.49344,70+0+0.49344)

Cluster 1

(00,0+0.19440,35,0+0.19440)

Time taken to build model (full training data) : 0 seconds

*** Model and evaluation on training set ***

Clustered Instances

0 3 (60%)

1 2 (40%)

Q2 - Answer

Cluster mode

☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
(Doc) Document
☒ More clusters for visualization

Ignore attributes

Start Stop

Result for (right-click for options)

06-51-49 - SimpleKMeans

06-51-50 - SimpleKMeans

06-52-07 - SimpleKMeans

Cluster output

KMeans

Random of iterations: 2

Within cluster sum of squared errors: 0.179209104590902

Initial starting points (random):

Cluster 0: 0.508170,0.3,0.1

Cluster 1: 0,1,0,0

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster0	Cluster1
	(3,0)	02,00	13,01
Word0	0.4415	0.5483	0.0592
Word2	0,0	0,00	0,0
Word3	0,00	0,0	0.0000

Time taken to build model (full training data) : 0 seconds

*** Model and evaluation on training set ***

Clustered Instances

0 2 (60%)

1 3 (40%)

Class attribute: Document

Cluster mode

☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 36.5140 - SimpleKMeans
- 36.5138 - SimpleKMeans
- 36.5207 - SimpleKMeans
- 36.5325 - HierarchicalCluster

Cluster output

Test mode: Choose to cluster evaluation on training data

==== Clustering model (full training set) ===

Cluster 0
 (0.0, 0.0, 24574, 0.0, 0.0, 24070, 0.0, 0.0, 24070)

Cluster 1
 (0.0, 0.0, 12288, 0.0, 0.0, 12288)

Time taken to build model (full training data) : 0 seconds

==== Model and evaluation on training set ===

Clustered Instances

Cluster	Count	Percentage
0	2	40%
1	2	40%

Class attributes: Document
 Classes to clusters:

Class	Cluster
0	0
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
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80	0
81	0
82	0
83	0
84	0
85	0
86	0
87	0
88	0
89	0
90	0
91	0
92	0
93	0
94	0
95	0
96	0
97	0
98	0
99	0

Incorrectly clustered instances : 0.0 40 4

Status
 OK

Log

Q3 - Answer

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose SimpleKMeans -init0-max-candidates 100-periodic-pruning 10000-min-seeds 2.0-t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance-RForest-lar" -I 500 -num-iter 1000

Cluster mode

☒ Use training set
☐ Supplied test set
☐ Percentage split
☐ Classes to clusters evaluation
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

- 36.5140 - SimpleKMeans

Cluster output

Number of iterations: 6
 Within cluster sum of squared errors: 0.2402445759368771

Initial starting points (random):

Cluster 0: 0.615385, 89.3
 Cluster 1: 1.89.25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full data	Cluster 0	Cluster 1
Latitude	0.5385	0.1154	0.8205
Longitude	80.249	80.21	80.2733

Time taken to build model (full training data) : 0.81 seconds

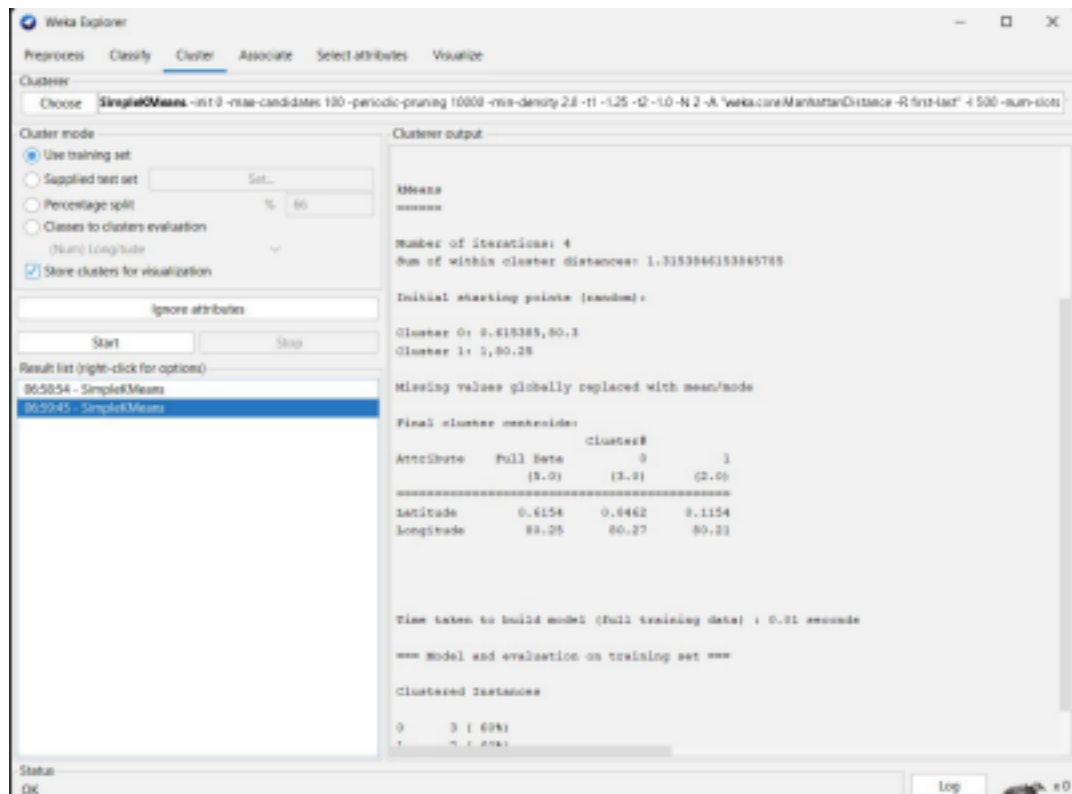
==== Model and evaluation on training set ===

Clustered Instances

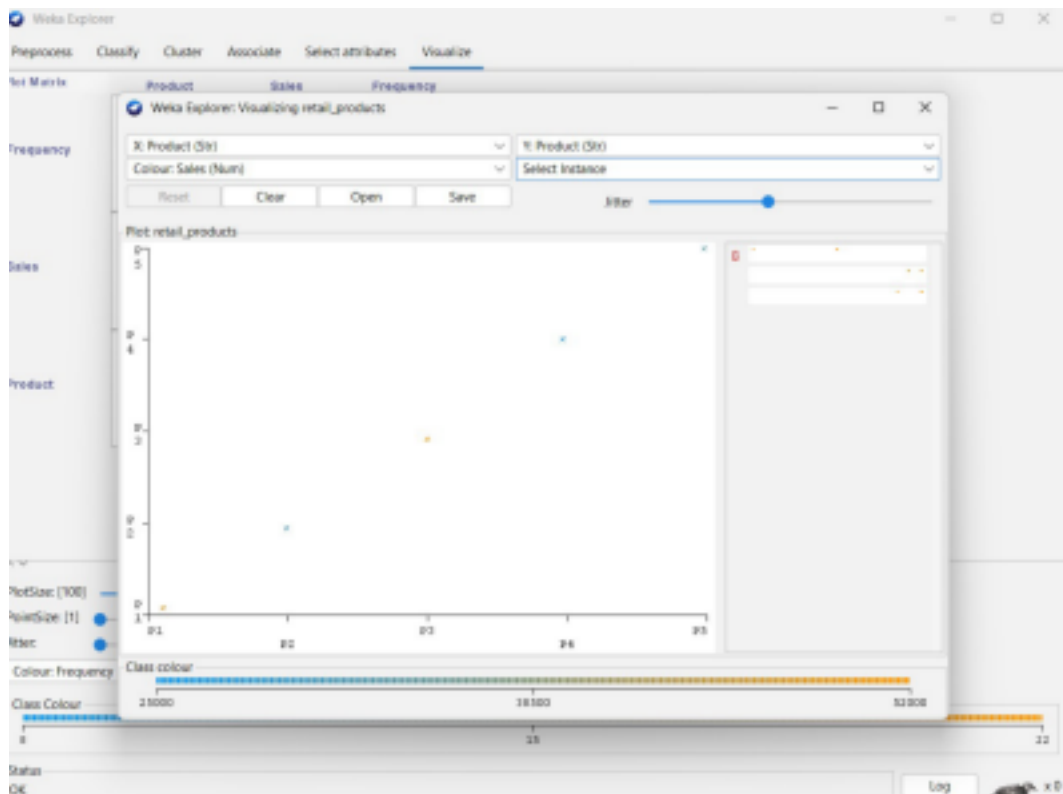
Cluster	Count	Percentage
0	2	40%
1	2	40%

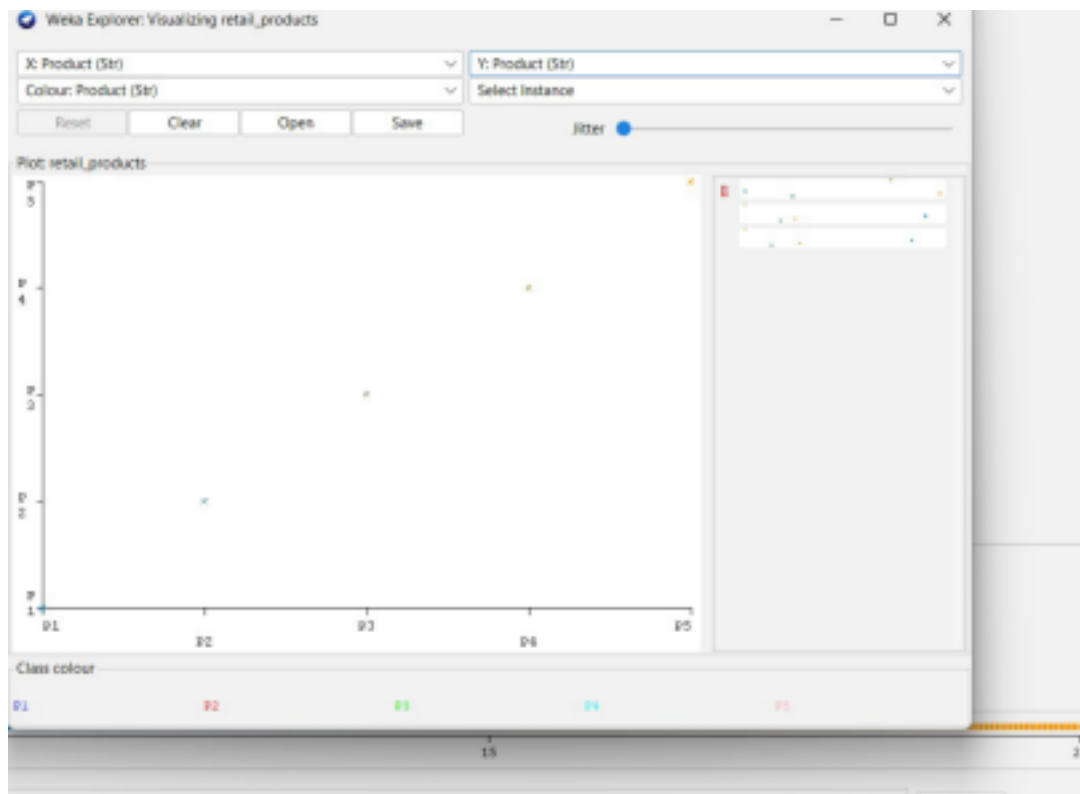
Status
 OK

Log



Q4 - Answer





Q5 - Answer

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster Choose **SimpleKMeans** - init 0 - max-candidates 100 - periodic-pruning 10000 - min-density 2.0 - t1 -1.25 - t2 -1.0 - N 2 - A "weka.core.EuclideanDistance" - R first-last - 4 500 - num-slots 1

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split % 66

☐ Classes to clusters evaluation (Num) Rating

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

00:11:18 - SimpleKMeans

Cluster output

=====
 Relation: mixed_attributes=weka.filters.unsupervised.attribute.Remove-R1=weka.filters.
 Instances: 5
 Attributes: 3
 Price
 Category#8
 Rating
 Test mode: evaluate on training data

=== Clustering model (full training set) ===

Means
 =====

Number of iterations: 3
 Within cluster sum of squared errors: 0.0751038518518518

Initial starting points (random):
 Cluster 0: 250,1,3,5
 Cluster 1: 300,1,3,0

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Cluster#		
	Full Data	0	1
	(5.0)	(2.0)	(3.0)
Price	424	275	523.3333
Category#8	0.4	1	0
Rating	4.22	3.65	4.6

Status OK Log